

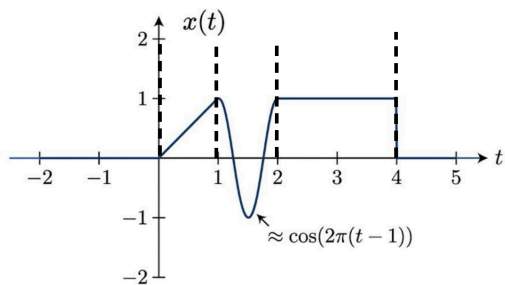
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TAREA 3. Representación Matemática de Señales.

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Para $x(t)$ obtener:



1) Representación matemática

$$2) \frac{d}{dt} x(t)$$

$$3) \int_{-\infty}^t x(\tau) d\tau$$

1) Representaciones Matemáticas

Inspección por recorrido:

$$x(t) = 0 + r(t) - r(t-1) + (\cos(2\pi(t-1)) - 1)u(t-1) - (\cos(2\pi(t-1)) - 1)u(t-2) - u(t-4)$$

Ventanas

$$x(t) = 0 [u(t+\infty) - u(t)] + r(t) [1 - u(t-1)] + (\cos(2\pi(t-1)) - 1) [u(t-1) - u(t-2)] + 1 [u(t-2) - u(t-4)]$$

2) Derivada

$$\frac{d}{dt} x(t) = r(t) - r(t-1) + (\cos(2\pi(t-1)) - 1)u(t-1) - (\cos(2\pi(t-1)) - 1)u(t-2) - u(t-4)$$

$$\frac{d}{dt} r(t) - r(t-1) = u(t) - u(t-1) +$$

$$\frac{d}{dt} [\cos(2\pi(t-1)) - 1] u(t-1) = [\cos(2\pi(t-1)) - 1] \delta(t-1) - 2\pi \sin(2\pi(t-1)) u(t-1) +$$

$$\frac{d}{dt} [-\cos(2\pi(t-1)) - 1] u(t-2) = [-\cos(2\pi(t-1)) - 1] \delta(t-2) + 2\pi \sin(2\pi(t-1)) u(t-2) +$$

$$\frac{d}{dt} [-u(t-4)] = -\delta(t-4) =$$

$$u(t) - u(t-1) - 2\pi \sin(2\pi(t-1)) [u(t-1) - u(t-2)] - \delta(t-4)$$

3) Integral

$$\int_{-\infty}^t x(\tau) d\tau = \int_0^t r(\tau) [1 - u(\tau-1)] d\tau + \int_1^t \cos(2\pi(\tau-1)) d\tau + \int_2^t 1 d\tau + \int_4^t 1 d\tau$$

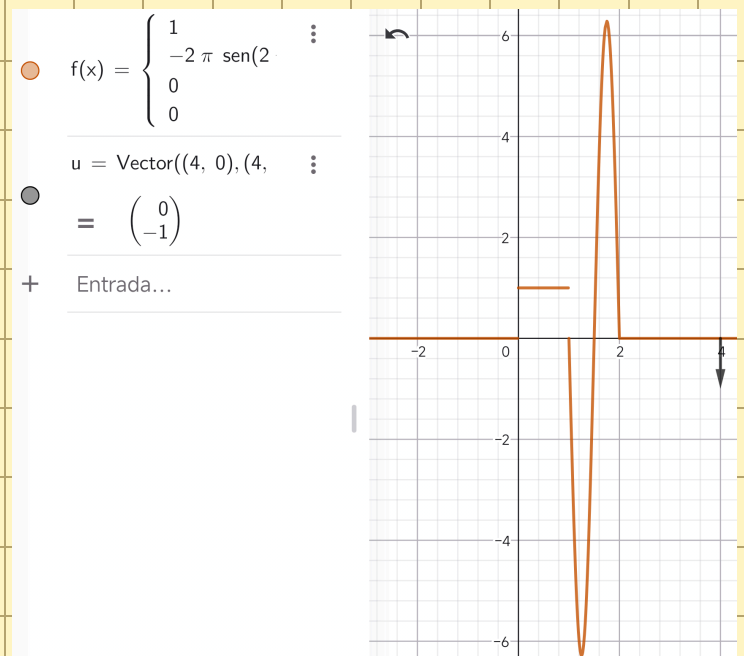
$$\int_0^t r(\tau) [1 - u(\tau-1)] d\tau = \frac{t^2}{2} [u(t) - u(t-1)] + \frac{1}{2} u(t-1)$$

$$\int_1^t \cos(2\pi(\tau-1)) d\tau = \left[\frac{\sin(2\pi(\tau-1))}{2\pi} \right]_1^t = \frac{\sin(2\pi(t-1))}{2\pi} = 0$$

$$\int_2^t 1 d\tau = (t-2) \quad \int_4^t 1 d\tau = (t-4) [u(t-2) - u(t-4)] + 2u(t-4)$$

$$Y(t) = \int_{-\infty}^t x(\tau) d\tau = \frac{t^2}{2} [u(t) - u(t-1)] + \left(\frac{1}{2} + \frac{\sin(2\pi(t-1))}{2\pi} \right) [u(t-1) - u(t-2)] + (t - \frac{3}{2}) [u(t-2) - u(t-4)] + \frac{5}{2} u(t-4)$$

Gráfica de la derivada



Gráfica de la integral

